Project Planning Phase — Project Milestones & Tasks

Date	_____
Team ID	_____
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis

1. Project Timeline Overview

The project is structured across 8 weeks with 8 clearly defined milestones. Each milestone represents a significant, verifiable deliverable that the team must complete before moving to the next. The timeline follows an Agile approach with 4 sprints nested within the 8-week framework.

2. Milestone Table

MS No.	Week	Milestone	Key Tasks	Owner	Status
MS-1	Week 1	**Project Setup & Data Collection**	Download Kaggle EV dataset; review schema and columns; set up GitHub repo; assign team roles; install Python and Node.js development environment	Data Eng.	___
MS-2	Week 2	**Data Cleaning & Preprocessing**	Remove null values; normalise all units (km, °C, kWh); engineer derived features (efficiency, degradation index, season); export clean CSV; validate output	Data Eng.	___
MS-3	Week 3	**Backend API Development**	Build Flask REST API; create /api/ev-data endpoint with all filter parameters; test with Postman; deploy to Heroku; document all endpoints	Backend Dev	___
MS-4	Week 4	**Core Visualisations Build (Sprint 2)**	Build Range Heatmap, Charge Curve, Model Comparison Bar Chart, Temperature Scatter Plot using D3.js and Plotly; connect each chart to live API	Viz Dev	___
MS-5	Week 5	**Filters and Interactivity (Sprint 3)**	Build filter panel with model, season, temperature, terrain, and date controls; implement cross-chart linking; add tooltips and plain-English annotations to all charts	Frontend Dev	___
MS-6	Week 6	**Degradation Chart and Summary Stats**	Build Battery Degradation Trend Line chart; build Summary Stats KPI panel; test all filter combinations across all charts	Viz Dev + Tester	___

MS-7	Week 7	**Export, Share, and Polish (Sprint 4)**	Build PNG download and PDF export; implement shareable URL with encoded filter state; fix all bugs found in testing; apply final visual polish	Frontend Dev	___
MS-8	Week 8	**Tableau Story and Final Documentation**	Build 10-slide Tableau Public story; publish dashboard and story publicly; write all project documents; complete final report; prepare demonstration	All Team	___

3. Detailed Task Breakdown by Milestone

MS-1: Project Setup & Data Collection (Week 1)

- Download the Kaggle EV Dataset (whc-ev-dataset.csv) and review all columns and data types.
- Create a shared GitHub repository with the agreed folder structure: /data, /src, /docs, /tests.
- Install and configure Python 3.10, Node.js 18, VS Code, and all required dependencies.
- Assign team roles (Data Engineer, Backend Developer, Visualisation Developer, Frontend Developer, Tester, Documentation Lead).
- Review the problem statement and confirm the scope of data available for the project.

MS-2: Data Cleaning & Preprocessing (Week 2)

- Identify and remove all rows with null values in key columns (model, range_km, battery_kwh, temp_celsius).
- Normalise range to km, temperature to °C, and battery capacity to kWh where mixed units exist.
- Engineer derived features: range_efficiency (range / battery_kwh), degradation_index (range vs. new spec), season label (Spring / Summer / Autumn / Winter from session_date).
- Export the cleaned DataFrame as clean_ev_data.csv and validate with a data quality check script.
- Document all preprocessing decisions in the Data Preparation document (P5_02).

MS-3: Backend API (Week 3)

- Build Flask REST API with three endpoints: /api/ev-data, /api/models, /api/summary.
- Implement all filter parameters on /api/ev-data: model (multi-select), season, temp_min, temp_max, terrain, date range.
- Add Flask-CORS to allow the React frontend to call the API from a different origin.
- Test all endpoints with Postman; confirm JSON responses match expected schema.
- Deploy Flask API to Heroku; confirm public access.

MS-4 to MS-8: Refer to Sprint Delivery Plan (P4_02)

Detailed task breakdowns for Weeks 4–8 are documented in the Sprint Delivery Plan (P4_02), which includes user story assignments, story point estimates, and sprint review criteria for Sprints 2, 3, and 4.